

module2day4

☒ OSPF

1. ☆ What is OSPF?

OSPF = Open Shortest Path First

- **IGP** → used inside an autonomous system.
- **Link-state routing protocol**
- **Classless** → supports VLSM/CIDR.
- Uses **SPF (Shortest Path First)** algorithm.
- Supports **authentication** and route summarization.
- **OSPFv2** → IPv4
- **OSPFv3** → IPv6

Remember



OSPF = Link-State + SPF + Cost + Areas



2. ☆ OSPF Main Components

LSA – Link-State Advertisement

Contains information about:

- Router links
- Networks
- Link costs

LSDB – Link-State Database

Stores the LSAs received by routers.

Routers in the **same OSPF area** maintain the same LSDB.

SPF / SPT

OSPF runs the **Shortest Path First algorithm** to calculate the best paths.

```
LSAs
↓
LSDB
↓
SPF Algorithm
↓
Shortest Path Tree
↓
Routing Table
```

Copy

☐ **Very important sequence!**





3. ☆ OSPF Areas

An **OSPF area** is a logical grouping of routers/interfaces.

Area ☐

Area 0 = Backbone Area

All normal non-backbone areas should connect to **Area 0**.

Example:



```
Area 1
|
|
[ ABR ]
|
Area 0
|
[ ABR ]
|
Area 2
```

ABR

ABR = Area Border Router

A router connected to:

- Area 0
- At least one other area

It exchanges routes between areas.

Why use multiple areas?

- Smaller LSDB
- Less LSA flooding
- Faster SPF calculations
- Better scalability
- Limits the impact of topology changes



4. ☆ OSPF Packet Types

MEMORIZE THESE 5!

Type	Packet	Purpose
1	Hello	Discover/maintain neighbors
2	DBD	Describe/summarize LSDB
3	LSR	Request missing LSAs
4	LSU	Send LSAs
5	LSAck	Acknowledge LSAs



Hello → DBD → LSR → LSU → ACK



5. ☆ OSPF Neighbor States

This is **very likely** to appear in an exam.

Correct order:

Down → Init → 2-Way → ExStart → Exchange → Loading → Full

① Down

No Hello received.

② Init

Hello received, but the router's own RID is not listed in the neighbor's Hello.

③ 2-Way

Two-way communication established.

On broadcast networks, **DR/BDR election occurs here**.

④ ExStart

Routers determine master/slave relationship for DBD exchange.

⑤ Exchange

Routers exchange DBD packets.

⑥ Loading

Missing LSAs are requested using LSRs.

⑦ Full

LSDB synchronization is complete. □

Memorize:



D → I → 2 → E → E → L → F



6. ☆ Router ID (RID)

OSPF needs a unique **Router ID**.

Cisco selection order:

1 Manually configured RID

```
router ospf 1  
router-id 1.1.1.1
```

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2 Highest active Loopback IP

3 Highest active physical interface IP

So:



Manual → Loopback → Physical

⚠ Router IDs must be **unique**.

If you change the RID, you may need:

```
clear ip ospf process
```

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7. ☆ OSPF Process ID

Example:

```
router ospf 1
```

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The **1** is the process ID.

Important:

The OSPF process ID is **locally significant**.

Therefore:

```
R1 → router ospf 1  
R2 → router ospf 5
```

[Copy](#)

They can still become OSPF neighbors.

☐ **Process IDs do NOT have to match.**



8. ☆ OSPF Neighbor Requirements

For neighbors to form an adjacency, check:

1. ☐ Unique Router IDs
2. ☐ Same subnet
3. ☐ Same Area ID
4. ☐ Same Hello timer

5. ☐ Same Dead timer
6. ☐ Compatible authentication
7. ☐ Compatible area type
8. ☐ Matching MTU

Most common exam answer:



Area ID + Hello/Dead timers + Authentication + MTU + subnet



9. ☆ OSPF DR & BDR

On **broadcast multi-access networks**, OSPF elects:

- **DR = Designated Router**
- **BDR = Backup Designated Router**

Why?

To reduce the number of OSPF adjacencies.

Without DR/BDR:

$$N(N-1) \frac{N(N-1)}{2}$$

With DR/BDR:

Election order:

1. Highest interface priority

2. Highest Router ID

Priority range:

0-255

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Important:

Priority 0 = cannot become DR/BDR



10. ☆ DR Election Is Non-Preemptive

This is a common tricky question. ⚠

If a router with a higher priority joins **after** the DR has already been elected:

× It does NOT automatically replace the current DR.

The election is **non-preemptive**.

A new election normally requires a failure/restart/event that causes the election process to occur again.





11. ☆ Point-to-Point Networks

Point-to-point OSPF networks:

× Do not elect DR/BDR.

Example:

R1 — R2

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Remember:



Broadcast LAN → DR/BDR



Point-to-Point → No DR/BDR



12. ☆ OSPF Metric / Cost

OSPF uses **cost** to select the best path.

Formula:

$$\text{Cost} = \frac{\text{Reference Bandwidth}}{\text{Interface Bandwidth}}$$

Lower total cost = **better path**.

Cisco's traditional default reference bandwidth is:

100 Mbps

⚠ This means FastEthernet and Gigabit Ethernet can both end up with cost 1 unless the reference bandwidth is changed.

Change reference bandwidth:

```
router ospf 1
auto-cost reference-bandwidth 1000
```

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Set interface cost directly:

```
interface g0/0
ip ospf cost 10
```

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13. ☆ OSPF Basic Configuration

Method 1 – Network statement

```
router ospf 1
router-id 1.1.1.1
network 192.168.10.0 0.0.0.255 area 0
```

[Copy](#)

Method 2 – Interface-specific

```
interface g0/0
ip ospf 1 area 0
```

[Copy](#)

- ☐ Interface-specific OSPF configuration takes precedence over network statements.



14. ☆ Wildcard Mask

OSPF network statements use a **wildcard mask**, not a subnet mask.

Example:

```
Subnet mask:
255.255.255.0
```

```
Wildcard:
0.0.0.255
```

[Copy](#)

Easy rule:



Wildcard = inverse of subnet mask



15. ☆ Passive Interface

```
passive-interface g0/1
```

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Prevents OSPF Hello packets from being sent out that interface.

Useful when an interface connects to an end device/network where no OSPF neighbor is expected.

Default all interfaces passive:

```
passive-interface default
```

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Then selectively enable OSPF neighbor formation on required interfaces.



16. ☆ Default Route Advertisement

```
default-information originate
```



Advertises a default route into OSPF **if a default route exists in the local routing table**.

To advertise it even when the router doesn't currently have a default route:

```
default-information originate always
```



Remember:



originate = advertise default



always = advertise even without local default route



17. ☆ OSPF Route Types

Code	Meaning
O	Intra-area
O IA	Inter-area
O E1	External Type 1
O E2	External Type 2

☐

Route learned within the same area.

☐ IA

Route learned from another OSPF area through an ABR.

E1/E2

External routes redistributed into OSPF.



18. ☆ OSPF LSA TYPES

This is **extremely important for exams**. ☐

Type	Name	Main Function
1	Router LSA	Describes router's links

Type	Name	Main Function
2	Network LSA	Created by DR
3	Summary LSA	ABR advertises networks between areas
4	ASBR Summary	ABR provides route to ASBR
5	AS External	ASBR advertises external routes
7	NSSA External	External routes inside NSSA

Easy memory:

1 = Router

2 = Network

3 = Summary

4 = ASBR

5 = External

7 = NSSA External

⚠ Type 6 is not part of the common IPv4 OSPF LSA list you need here.



19. ☆ OSPF Authentication

Three authentication methods:

1. **Null authentication** — default
2. **Simple password** — plaintext
3. **MD5** — cryptographic authentication

Exam point:

MD5 is more secure than plaintext authentication.

Authentication settings/credentials must be compatible for neighbors to form adjacency.



20. ☆ OSPF Timers

Two important timers:

Hello interval

How frequently Hello packets are sent.

```
ip ospf hello-interval 10
```

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Dead interval

How long the router waits before declaring a neighbor down.

```
ip ospf dead-interval 40
```

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□ On Cisco, the default relationship is commonly:

Dead=4×HelloDead = 4 \times Hello

Neighbors need compatible timer values.



21. ☆ OSPF Verification Commands

Check OSPF neighbors

```
show ip ospf neighbor
```

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Look for:

FULL

[Copy](#)

Check OSPF interfaces

```
show ip ospf interface brief
```

[Copy](#)

Check OSPF routes

```
show ip route ospf
```

[Copy](#)

Check LSDB

```
show ip ospf database
```

[Copy](#)

Check OSPF configuration/protocol information

```
show ip protocols
```

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22. ☆ Administrative Distance

Default OSPF Administrative Distance:

110

Remember:



OSPF = AD 110



23. ☆ OSPF Troubleshooting

If two routers are **not becoming FULL**, check these first:

1 Area

Area ID must match

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2 Hello/Dead timers

Must match

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3 Authentication

Must match

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4 MTU

Must match

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5 Subnet

Interfaces must share a common subnet

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6 Passive interface

Make sure the interface isn't unintentionally passive.

7 Router ID

Make sure RIDs are unique.



24. ☆ MTU Mismatch

A particularly important troubleshooting scenario:

Routers may discover each other but fail during database exchange because of an **MTU mismatch**.

Think:

Hello works



Neighbor discovered



DBD exchange fails



Adjacency does not reach FULL

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25. ☆ OSPF Filtering

istribute-list

Example:

```
distribute-list prefix MYLIST in
```

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Important concept:

A local OSPF distribute-list does NOT remove LSAs from the LSDB.

It mainly controls whether routes are installed into the **local RIB**.

Remember:



distribute-list = local routing-table filtering

Not:



× LSA flooding control



26. ☆ ABR Route Filtering

ABRs can filter inter-area prefixes using:

```
area <area-id> filter-list prefix <prefix-list-name> in
```

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or:

```
area <area-id> filter-list prefix <prefix-list-name> out
```

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area range

Used for inter-area route summarization.

area 1 range 192.168.0.0 255.255.0.0

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